

SOC ENGINEER PRACTICAL PROJECT

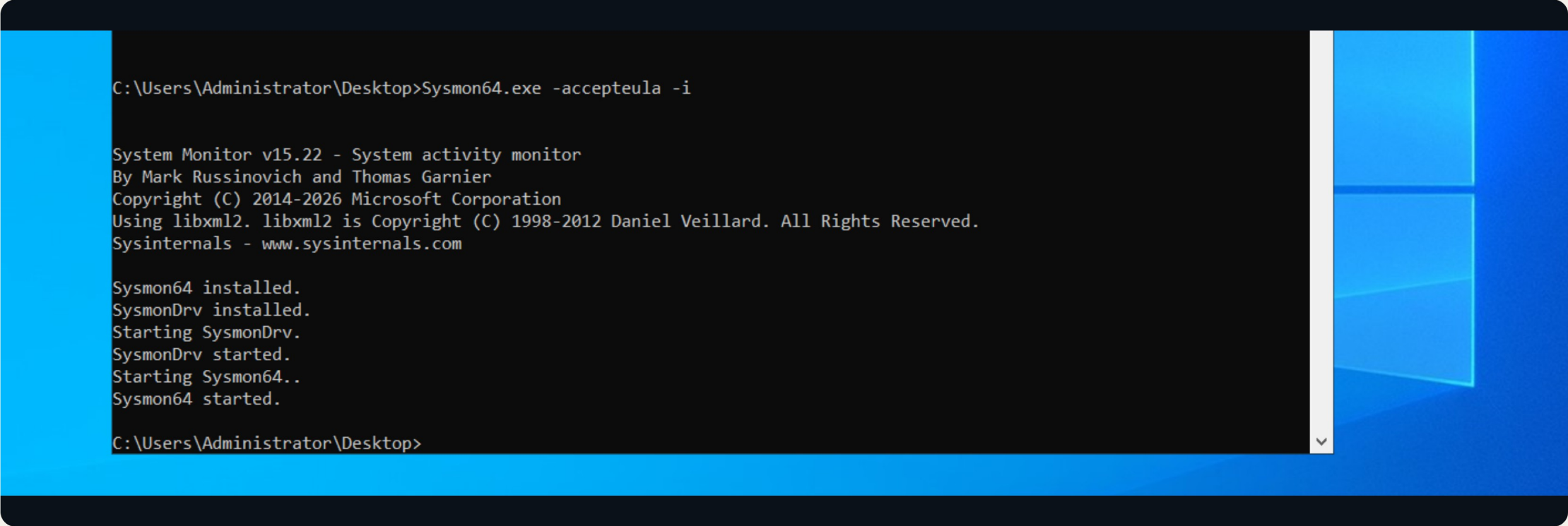
Task 2: Windows AD, PowerShell and Sysmon Logs to Splunk

Collecting endpoint telemetry and domain controller events for threat hunting

STEP 01 | SYSMON

Install Sysmon

Sysmon64 installed from the command line with the EULA accepted. Both the SysmonDrv driver and the Sysmon64 service started.



```
C:\Users\Administrator\Desktop>Sysmon64.exe -accepteula -i

System Monitor v15.22 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Sysmon64 installed.
SysmonDrv installed.
Starting SysmonDrv.
SysmonDrv started.
Starting Sysmon64..
Sysmon64 started.

C:\Users\Administrator\Desktop>
```

STEP 02 | SYSMON

Verify the Sysmon service

The sc query output shows the Sysmon64 service in the RUNNING state on the domain controller.

```
C:\Users\Administrator\Desktop>sc query Sysmon64

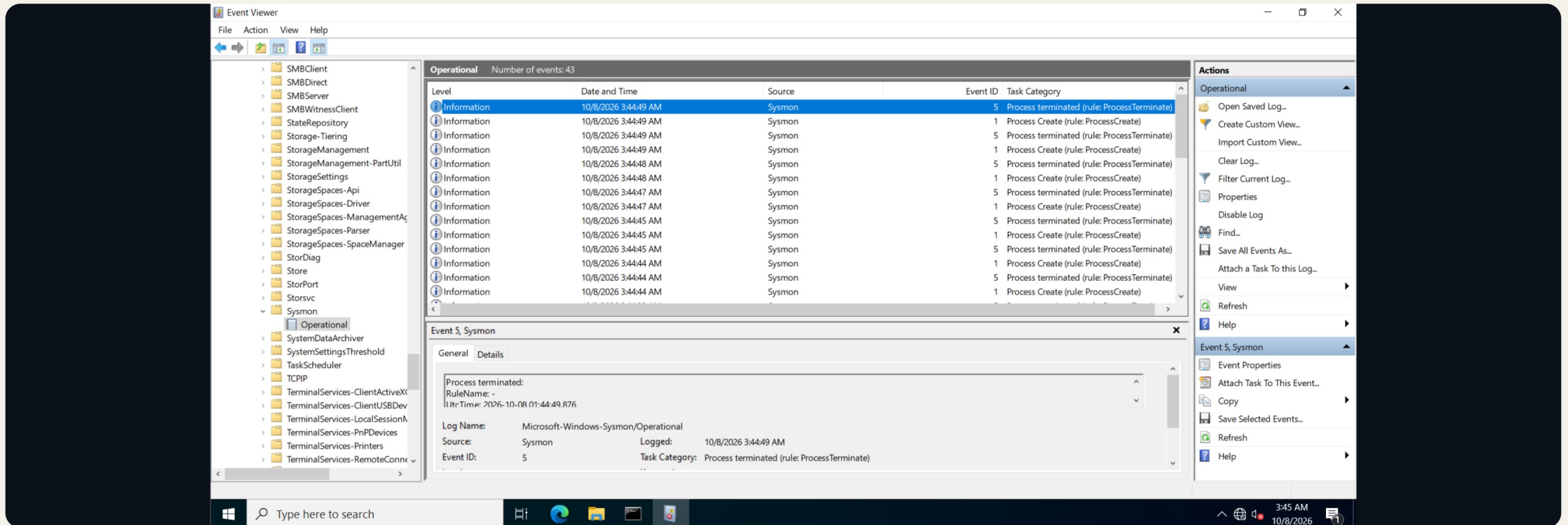
SERVICE_NAME: Sysmon64
        TYPE               : 10  WIN32_OWN_PROCESS
        STATE                : 4   RUNNING
                           (STOPPABLE, NOT_PAUSABLE, IGNORES_SHUTDOWN)
        WIN32_EXIT_CODE       : 0   (0x0)
        SERVICE_EXIT_CODE    : 0   (0x0)
        CHECKPOINT            : 0x0
        WAIT_HINT             : 0x0

C:\Users\Administrator\Desktop>
```

STEP 03 | SYSMON

Sysmon events in Event Viewer

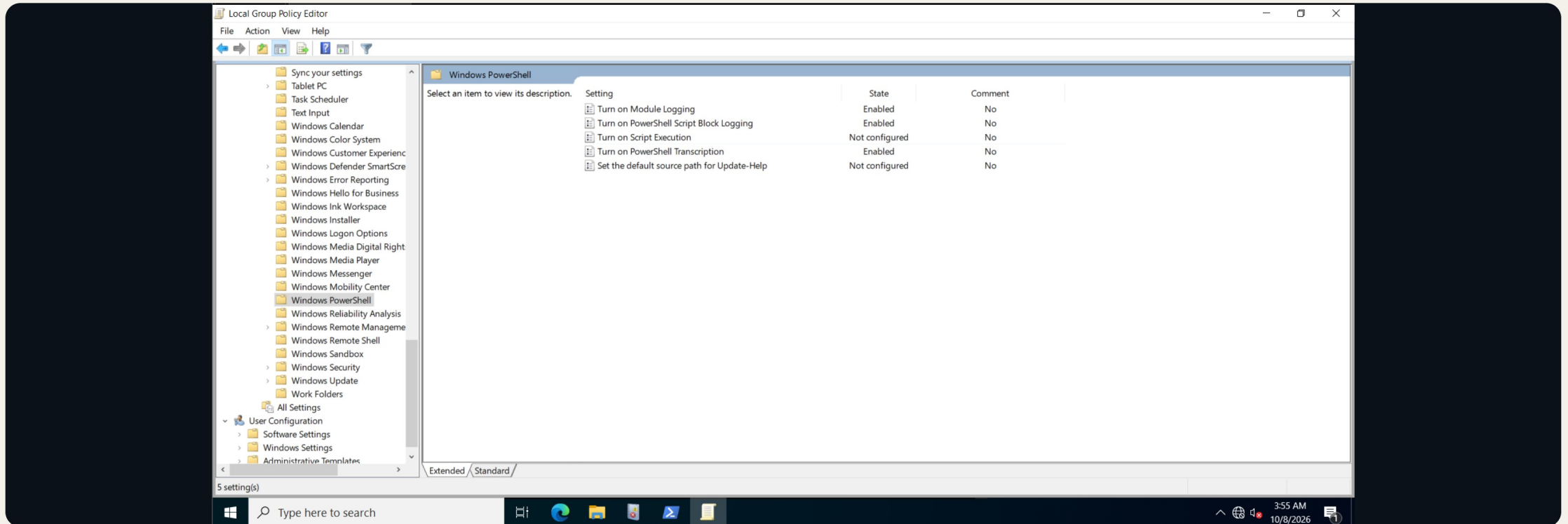
The Microsoft-Windows-Sysmon/Operational log is filling with Process Create (ID 1) and Process terminated (ID 5) events.



STEP 04 | POWERSHELL LOGGING

Group Policy settings

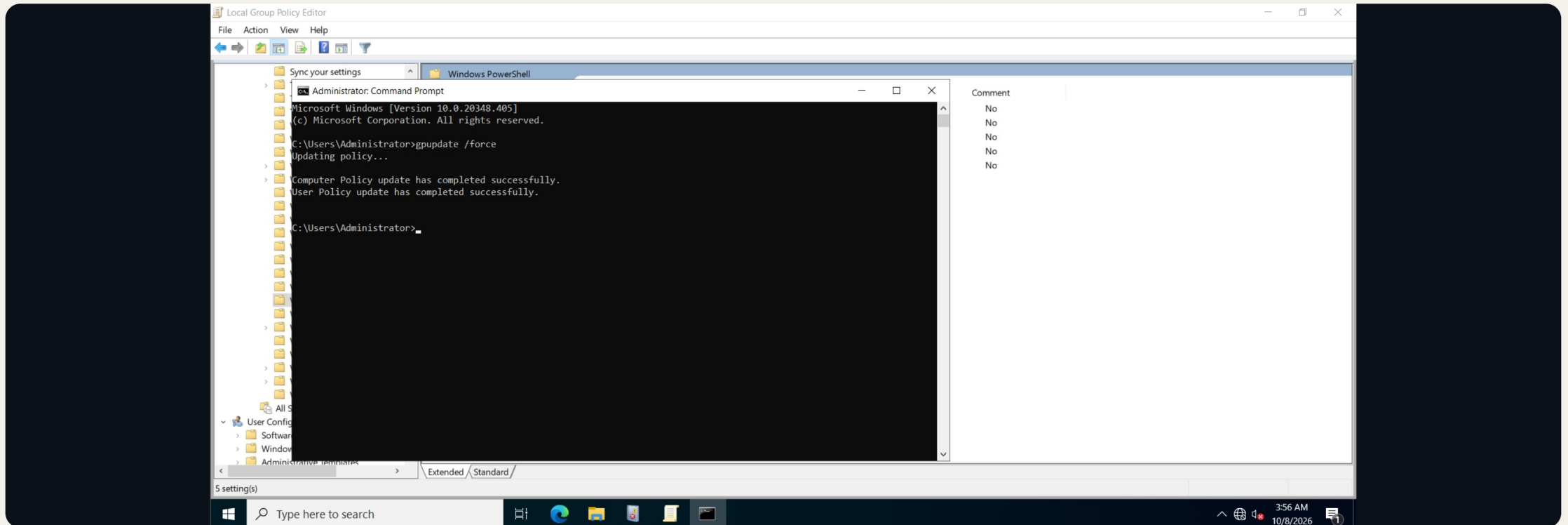
Module Logging, Script Block Logging and PowerShell Transcription are all set to Enabled in the Windows PowerShell policy folder.



STEP 05 | POWERSHELL LOGGING

Apply the policy

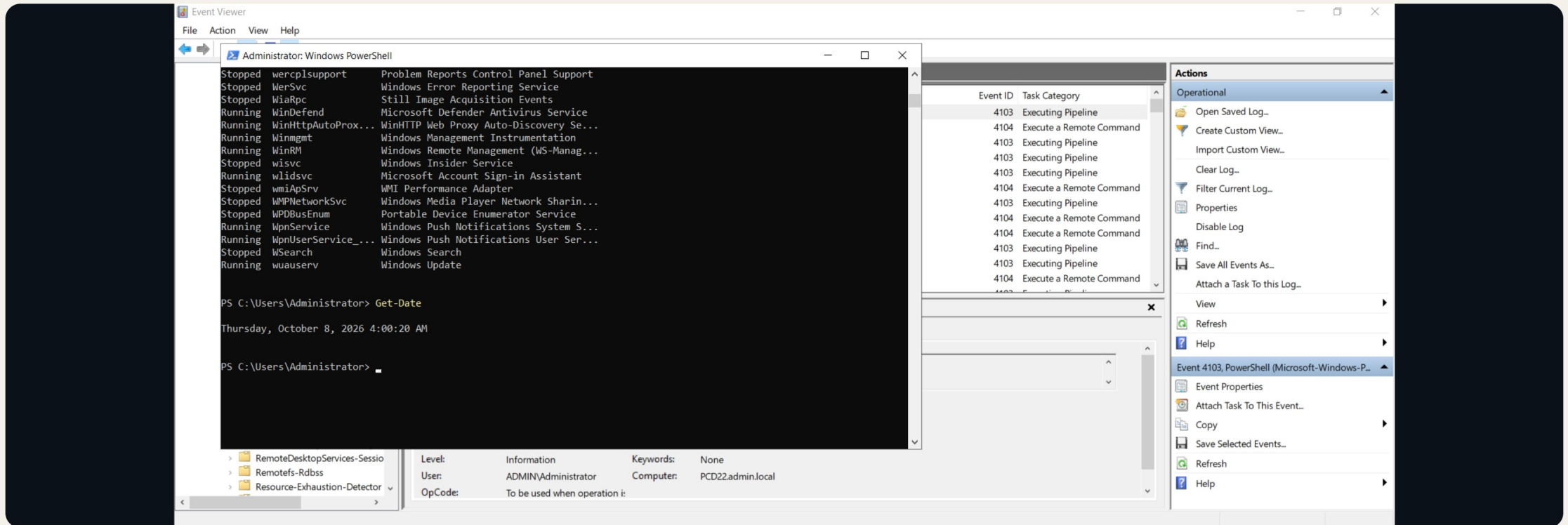
gpupdate /force completed successfully for both the computer and user policies.



STEP 06 | POWERSHELL LOGGING

Generate PowerShell activity

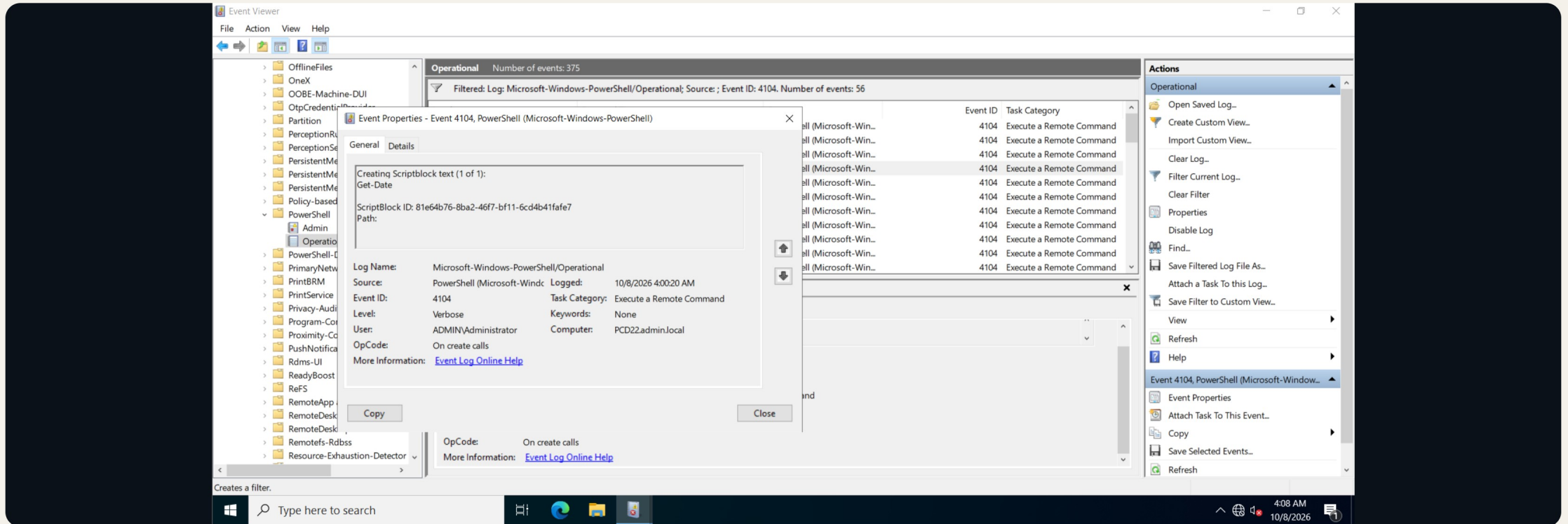
Commands run in an elevated PowerShell session, ending with Get-Date, to create a recognizable test event.



STEP 07 | POWERSHELL LOGGING

Event ID 4104 in Event Viewer

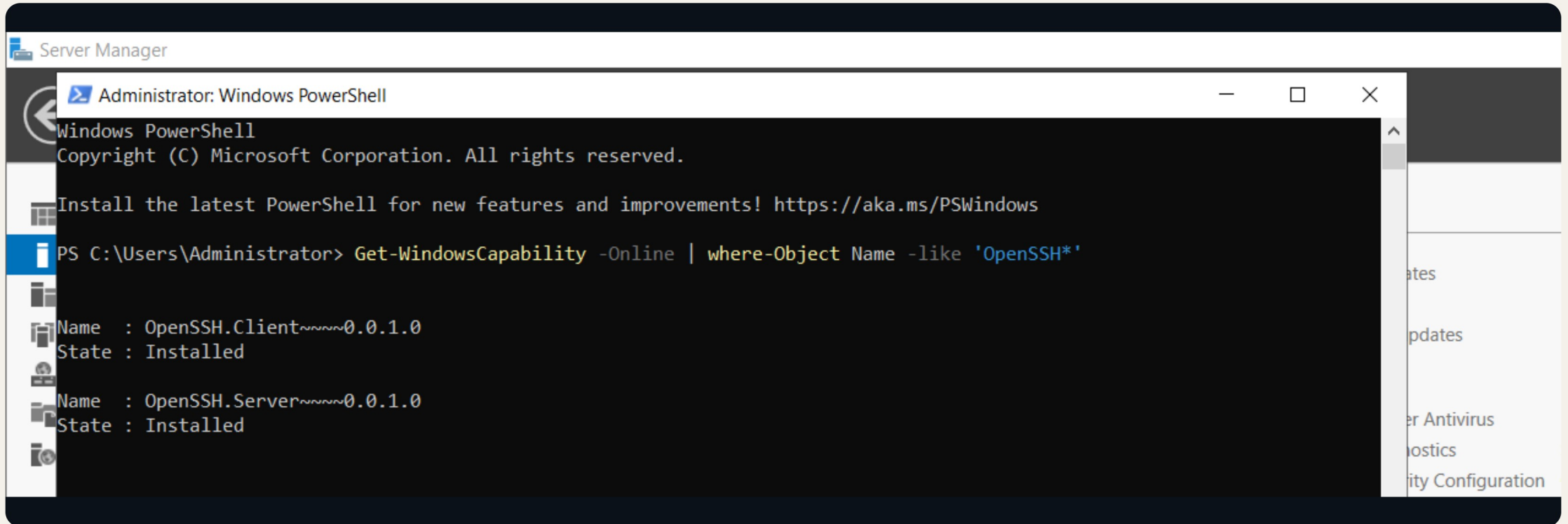
Script Block Logging recorded the Get-Date script block in the PowerShell/Operational log, with Event ID 4104.



STEP 08 | LOG FORWARDING

OpenSSH installed

OpenSSH Client and Server confirmed as installed on the Windows Server, to allow file transfer to the machine.



```
Server Manager

Administrator: Windows PowerShell

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Administrator> Get-WindowsCapability -Online | where-Object Name -like 'OpenSSH*'

Name : OpenSSH.Client~~~~~0.0.1.0
State : Installed

Name : OpenSSH.Server~~~~~0.0.1.0
State : Installed
```

Start the SSH service

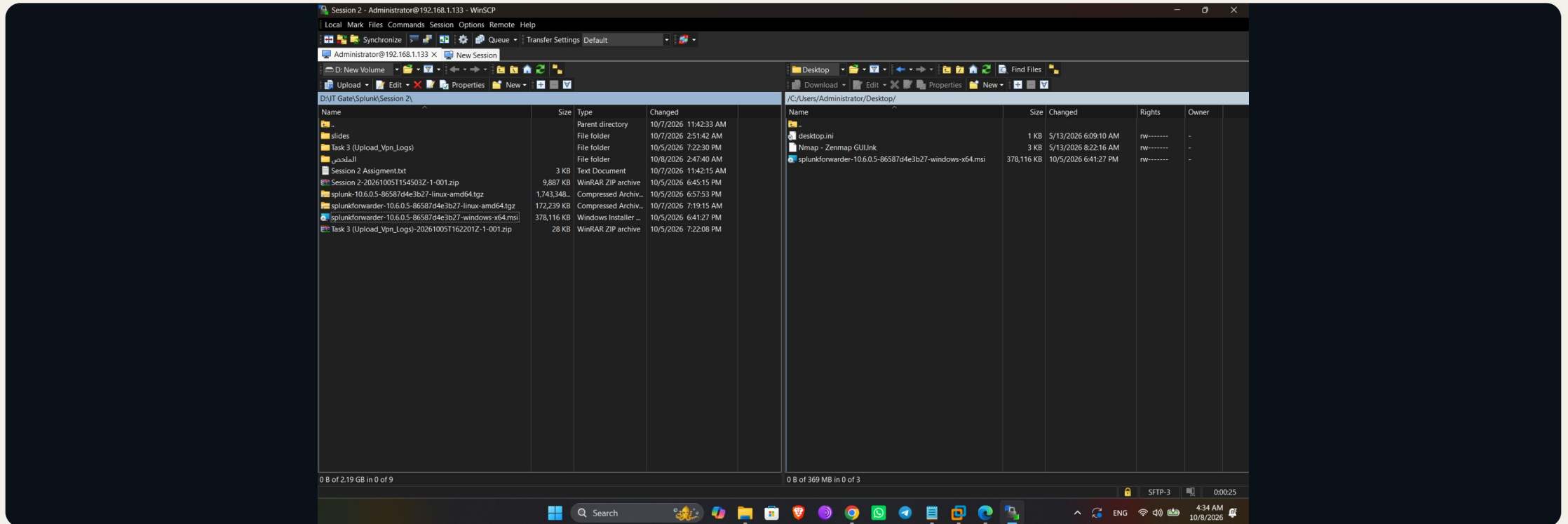
The sshd service started from PowerShell.

```
PS C:\Users\Administrator> Start-Service sshd
```

STEP 10 | LOG FORWARDING

Transfer the forwarder

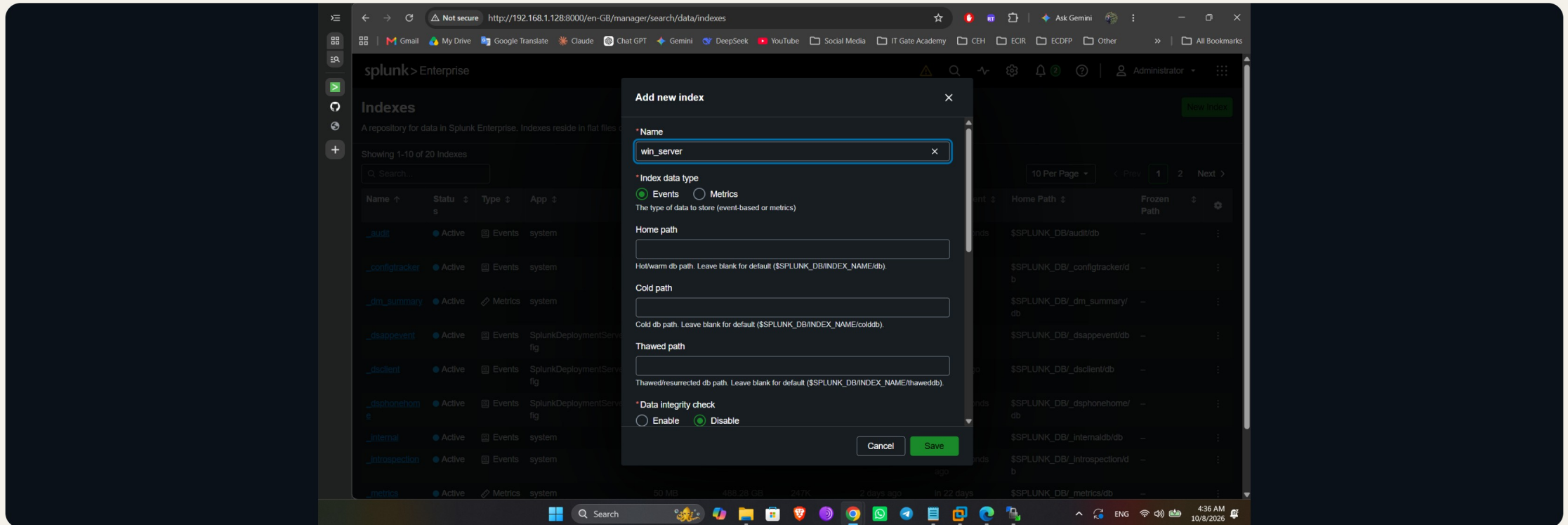
The Splunk Universal Forwarder MSI copied from the host PC to the server desktop using WinSCP over SFTP.



STEP 11 | LOG FORWARDING

Create the win_server index

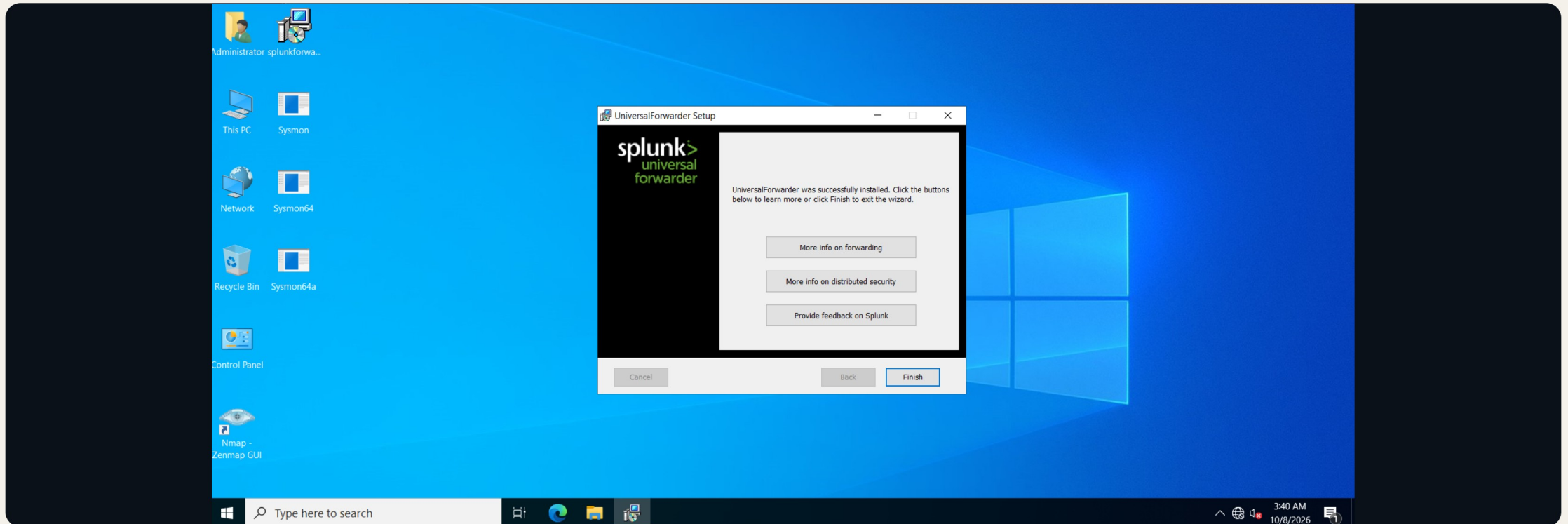
A dedicated events index named win_server created in Splunk Enterprise for all Windows telemetry.



STEP 12 | LOG FORWARDING

Install the Universal Forwarder

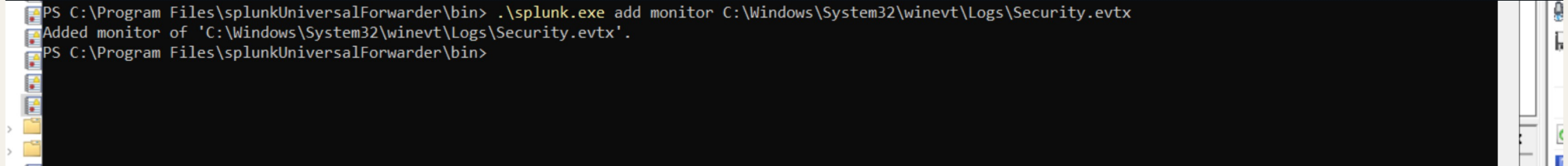
The Universal Forwarder installed successfully on the domain controller.



STEP 13 | LOG FORWARDING

Add the Security log monitor

The forwarder CLI used to add a monitor on the Security.evtx event log.

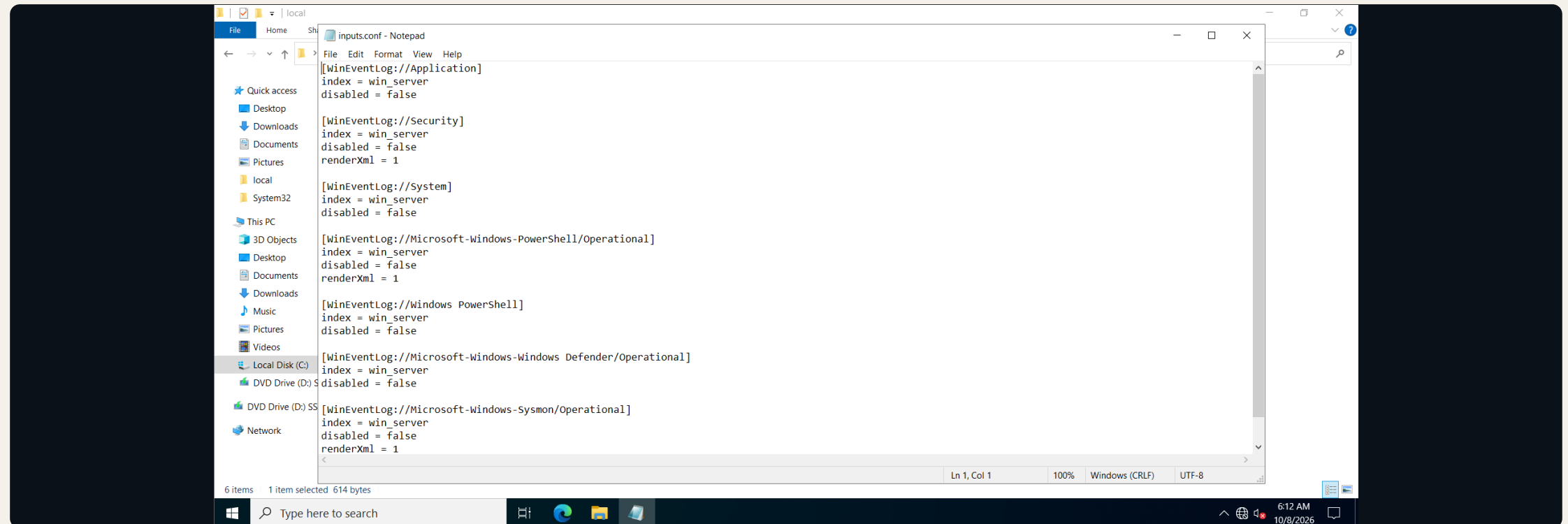


```
PS C:\Program Files\splunkUniversalForwarder\bin> .\splunk.exe add monitor C:\Windows\System32\winevt\Logs\Security.evtx
Added monitor of 'C:\Windows\System32\winevt\Logs\Security.evtx'.
PS C:\Program Files\splunkUniversalForwarder\bin>
```

STEP 14 | LOG FORWARDING

Configure inputs.conf

Stanzas for Application, Security, System, PowerShell, Windows PowerShell, Defender and Sysmon logs, all sent to the win_server index.



STEP 15 | SIEM INGESTION

Logs arriving in Splunk

15,052 events in win_server from seven sources, led by Security, System, Sysmon and PowerShell Operational.

New Search

index="win_server"

15,052 events (before 08/10/2026 05:43:42.000) No Event Sampling

Events (15,052) Patterns Statistics Visualization

Timeline format Zoom Out

7 Values, 100% of events

Values	Count	%
WinEventLog:Security	9,293	61.73%
WinEventLog:System	2,432	16.15%
WinEventLog:Microsoft-Windows-Sysmon/Operational	1,850	12.29%
WinEventLog:Microsoft-Windows-PowerShell/Operational	501	3.32%
WinEventLog:Application	499	3.31%
WinEventLog:Windows PowerShell	380	2.54%
WinEventLog:Microsoft-Windows-Defender/Operational	97	0.64%

Selected Yes No

Reports: Top values, Top values by time, Rare values

Events with this field

XML Event Data Preview:

```
<Event xmlns="http://schemas.microsoft.com/win/2004/08/events/event"><System><Provider Name="Microsoft-Windows-Security-Auditing" Guid="{54849625-5478-4994-a5ba-3e33-000000000000}"><Level>1</Level><Task>12644</Task><Opcode>0</Opcode><Keywords>0x8020000000000000</Keywords><TimeCreated SystemTime="2026-10-08T02:41:40.9688732Z"><EventRecordID>9292</EventRecordID><Correlation ActivityID="{29612932-56c3-0001-6429-6129c356d801}"><Execution ProcessID="684" ThreadID="732"><Channel>Security</Channel><Computer>PCD22.admin.local</Computer><Security/></System><EventData><Data Name="SubjectUserSid">S-1-5-18</Data><Data Name="SubjectUserName">PCD22</Data><Data Name="SubjectDomainName">ADMIN</Data><Data Name="SubjectLogonId">0x781d4b</Data><Data Name="PrivilegeList">SeSecurityPrivilege</Data></EventData></Event>
```

STEP 16 | SIEM INGESTION

PowerShell events in Splunk

Event Code 4104 events from the PowerShell Operational log, with the Get-Date script block visible in the Message field.

The screenshot displays the Splunk Search & Reporting interface. The left sidebar shows navigation options: Search & Reporting, Search, Datasets, Reports, Alerts, Dashboards, and Modules. The main panel shows a search results table with two events. The second event is selected, showing details for Event Code 4104. The Message field contains the text: "Creating Scriptblock text (1 of 1): Get-Date". Below the event details, an "Event Actions" table lists various fields and their values, including host (PCD22), source (WinEventLog:Microsoft Windows PowerShell/Operational), sourcetype (WinEventLog:Microsoft Windows PowerShell/Operational), ComputerName (PCD22.admin.local), EventCode (4104), EventType (5), Keywords (None), LogName (Microsoft Windows PowerShell/Operational), Message (Creating Scriptblock text (1 of 1): Get-Date), OpCode (On create calls), RecordNumber (337), ScriptBlock_ID (a9d07c-8d47-4c7f-b07e-a33c86bd535c), Sid (S-15-21-3534638701-468094028-796588408-500), SslType (0), SourceName (Microsoft Windows PowerShell), TaskCategory (Execute a Remote Command), Type (Verbose), and User (NOT_TRANSLATED).

Type	Field	Value	Actions
Selected	host	PCD22	
	source	WinEventLog:Microsoft Windows PowerShell/Operational	
	sourcetype	WinEventLog:Microsoft Windows PowerShell/Operational	
Event	ComputerName	PCD22.admin.local	
	EventCode	4104	
	EventType	5	
	Keywords	None	
	LogName	Microsoft Windows PowerShell/Operational	
	Message	Creating Scriptblock text (1 of 1): Get-Date	
	OpCode	On create calls	
	RecordNumber	337	
	ScriptBlock_ID	a9d07c-8d47-4c7f-b07e-a33c86bd535c	
	Sid	S-15-21-3534638701-468094028-796588408-500	
	SslType	0	
	SourceName	Microsoft Windows PowerShell	
	TaskCategory	Execute a Remote Command	
	Type	Verbose	
	User	NOT_TRANSLATED	

STEP 17 | SIEM INGESTION

Hunt for the script block

Search on EventCode=4104 and Message containing Get-Date returns the matching script block events.

The screenshot shows the Splunk Enterprise web interface. The search bar contains the query: `index="win_server" EventCode=4104 Message="*Get-Date"`. The search results show 4 events. The selected event is from 08/10/2026 at 05:00:20.414 AM. The event details are as follows:

Type	Field	Value	Actions
Selected	host	PCD22	
	source	WinEventLog:Microsoft-Windows-PowerShell/Operational	
	sourcetype	WinEventLog:Microsoft-Windows-PowerShell/Operational	
Event	ComputerName	PCD22.admin.local	
	EventCode	4104	
	EventType	5	
	Keywords	None	
	LogName	Microsoft-Windows-PowerShell/Operational	
	Message	Creating Scriptblock test (1 of 9): Get-Date ScriptBlock ID: 81e64b76-8ba2-46b7-bf11-6c04b41f9f7 Path:	
	OpCode	On create calls	
	RecordNumber	366	
	ScriptBlock_ID	81e64b76-8ba2-46b7-bf11-6c04b41f9f7	
	ScriptBlock	Get-Date	

Sysmon events in Splunk

The screenshot displays the Splunk Enterprise interface. The search bar contains the query: `index="win_server" source="WinEventLog:Microsoft-Windows-Sysmon/Operational"`. The search results show a timeline of events, with a specific event selected for details. The event details show a Sysmon event with a description of a process execution. The event code is 4, and the event type is 1. The event details also show the event's parent process and the user who executed the process.

Result

Capture

Sysmon process telemetry and PowerShell Script Block Logging enabled on the domain controller.

Forward

Universal Forwarder ships Sysmon, PowerShell and Security logs to the win_server index.

Hunt

Events searchable in Splunk by EventCode, including 4104 script blocks and Sysmon process events.